

Introduction

Unit 1

Multimedia

- Multi - many;much;multiple
- Medium
 - Means for distribution and presentation of information
 - Classification based on perception (text, audio, video) is appropriate for defining multimedia
- Multimedia is a combination of text, graphics, sound, animation and video that is delivered interactively to the user by electronic or digitally manipulated means.

Multimedia

- Multimedia is the field concerned with the computer controlled integration text, graphics, drawings, still and moving image (video), animation, audio any other media where every type of information can be represented, stored, transmitted and processed digitally.

Building Blocks

- Text
 - Text and symbols are very important for communication.
 - With recent explosion of Internet and WWW, text has become more the important than ever. Web is HTML originally designed to display simple text document on screens.
- Audio
 - Produced by vibration, as perceived by the sense of hearing
 - In multimedia, audio could come in the form of speech effects and also music.

Building Blocks

- Graphics
 - Two dimensional figure or illustration.
 - Could be produced manually (drawing, painting, carving, etc) or by computer graphics technology
 - Used in multimedia to show more clearly what a particular information is all about (diagrams, ,picture)
- Video
 - Is the technology of capturing, recording, processing, transmitting and reconstructing moving pictures
 - Video also takes a lot of storage spaces. So plan carefully before you are going to use it.

Building Blocks

- Animation
 - Animation is the rapid display of a sequence of images of 2D artwork or model positions in order to create an illusion of movement.
 - It is an optical illusion of motion due to the phenomenon of persistence of vision.

History

1. Early Beginning

- The term "multimedia" was first coined by artist Bob Goldstein to describe his show "Lightworks at L'Oursin" in 1966.

2. 1970s-1980s

- 1972: The video game "Pong" by Atari is considered one of the earliest examples of multimedia, combining graphics and sound
- 1976: Steve Jobs and Steve Wozniak founded Apple Computer, and in 1977, they released the Apple II, the first computer to use color graphics
- 1981: IBM introduced its first PC, and in 1984, Apple released the Macintosh, which featured a graphical user interface (GUI) and a mouse

History

3. 1990s

- The multimedia was the probably one of the most overused term in 90s.
- The field was at the crossroads of several major industries: computing, telecommunications, publishing, consumer audio-video, electronics, and television/movie/broadcasting.



4. 2000s-present

- The rise of the internet and streaming services revolutionized multimedia, making it more accessible and interactive
- Present: Multimedia continues to evolve with advancements in virtual reality, augmented reality, and interactive media, providing more immersive experiences.

Application

1. WWW
2. Education
3. Medicine
4. Communication
5. Broadcasting
6. Entertainments and Games
7. Industry
8. Commercial

Multimedia System

- A Multimedia System is a system capable of processing multimedia data and applications.
- A Multimedia System is characterised by the processing, storage, generation, manipulation of Multimedia information.

Multimedia Application

A Multimedia Application is an application which uses a collection of multiple media sources e.g. text, graphics, images, sound/audio, animation and/or video

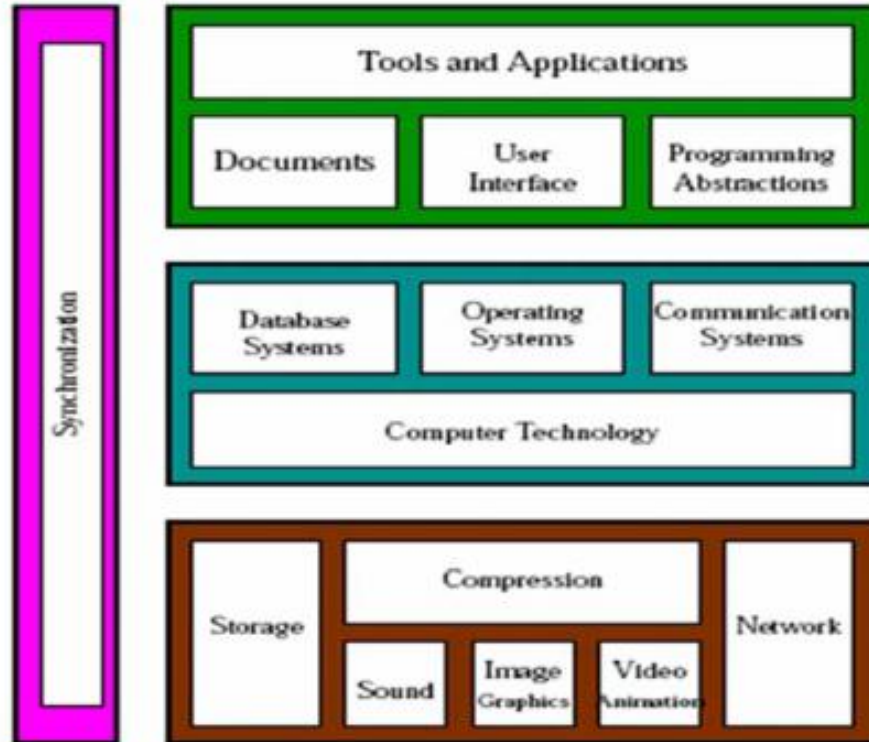
Example: (For students)

Global Structure of Multimedia

The global structure of multimedia is a comprehensive framework that encompasses the devices, systems, applications, and interactions required to capture, process, store, transmit, and present various types of media in a cohesive and interactive manner.

- Application Domain
- System Domain
- Device Domain

Global Structure of Multimedia



Application Domain

- Provides functions to the user to develop and present **multimedia projects**.
- This includes software tools, and multimedia projects development methodology.
- The services of the system domain are offered to the application domain through proper programming abstractions.

System Domain

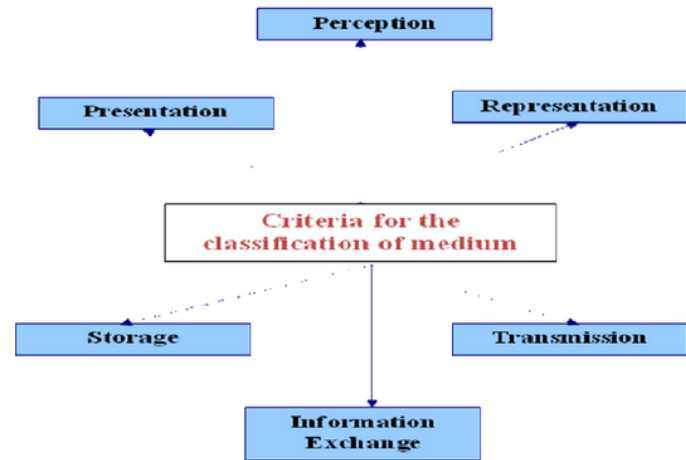
System domain — including **all supports for using the functions** of the device domain, e.g., operating systems, communication systems(networking) and database systems

Device Domain

- Basic concepts and skill for processing various multimedia elements and for handling physical device.
- This includes the hardware components like cameras, microphones, speakers, and other input/output devices that capture and present multimedia content.

Medium

- Medium is defined as means for distribution and presentation of information.
- Examples of a medium are text, graphics, speech, and music.
Media can be classified with respect to different criteria.
- We classify media according to
 - Perception Medium
 - Representation Medium,
 - Presentation Medium
 - Storage Medium
 - Transmission Medium
 - information exchange Medium



Perception Medium

- Perception media help human to sense their environment.
- The central question is: How human perceive information in a computer environment?
→The answer is through seeing and hearing.
- For the perception of information through **seeing** the usual such as text, image and video are used
- For the perception of information through **hearing** media such as music, noise and speech are used.

Representation Medium

- Representation media are defined by internal computer representation of information.
- The central question is: How the computer information is coded?
- The answer is that various formats are used to represent media information in a computer
 - Text, character is coded in ASCII code.
 - Graphics are coded according to CEPT or CAPTAIN video text standard.
 - Image can be stored as JPEG format
 - Audio video sequence can be coded in different TV standard format (PAL, NTSC, SECAM and stored in the computer in MPEG format).

Presentation Medium

- Presentation media refer to the tools and devices for the input and output of the information.
- The central question is: Through which the information is delivered by the computer and is introduced to the computer?
- The answer is:

Output media:

Paper, screen and speaker are the output media.

Input media:

Keyboard, mouse, camera, microphone are the input media.

Storage Medium

- Storage media refer to the data carrier which enables storage of information.
- The central question is: How will information be stored? The answer is hard disk, CD-ROM, etc.

Transmission Medium

- Transmission media are the different information carrier that enables continuous data transmission.
- The central question is: Over what will the information be transmitted?
- The answer is co-axial cable and fiber optics, as well as free air space transmission, which is used for wireless traffic.

Information Exchange Medium

- Information exchange media includes all information carrier for transmission, i.e. all storage and transmission media.
- The central question is: Which information carrier will be used for information exchange between different places?
- The answer is combine uses of storage and transmission media.
For example: Electronic Mailing System.

Media Concepts

- Each Media defines:
 1. Representation Spaces
 2. Representation Values
 3. Representation Dimensions

Representation Space

Representation space determines the surrounding where the media are presented.

- Visual representation space (e.g. paper, screen)
- Acoustic representation space (e.g. stereo)

Representation Values

Representation values – determine the information representation of different media

- Continuous representation values (e.g. electro-magnetic waves)
- Discrete representation values (e.g. text characters in digital form)

Representation Dimensions

- Representation dimensions of a representation space are:
 - Spatial Dimension
 - Temporal Dimension
- Spatial Dimension
 - Two dimensional (Monitor)
 - Three dimensional (holography)
- Temporal Dimension
 - Time independent - Discrete Media
 - Time Dependent - Continuous Media

Temporal Dimension

- Time independent (document) – Discrete media

Information consists of a sequence of individual elements without a time component.

- Time dependent (movie) – Continuous media

Information is expressed not only by its individual value but also by its time of occurrence.

Properties of Multimedia System

The uses of term multimedia are not every arbitrary combination of media. The main properties of multimedia system are:

- Combination of media
- Indepence
- Computer support Integration

Combination of Media

A simple text processing program with in corporate image is often called a multimedia application because two media are processed through one program. But one should talk multimedia only when both continuous and discrete media are utilized. So text processing program with incorporated images is not a multimedia application.

Independence

An important aspect of different media is their level of independence from each other. In general there is a request for independence of different media but multimedia may requires several level of independence. E.g. A computer controlled video recorder stores audio and video information's. There is inherently tight connection between two types of media. Both media are coupled together through common storage medium. On the other hand for the purpose of presentation the combination of DAT (digital audio tape recorder) signals and computer available text satisfies the request for media independence.

Computer Support Integration

Computer is idle tools for media combinations. The system should be capable of computer controlled media processing. The system should be programmable by a system programmer or even a user.

Characteristics of Multimedia System

A Multimedia system has four basic characteristics:

- Multimedia systems must be computer controlled.
- Multimedia systems are integrated.
- The information they handle must be represented digitally.
- The interface to the final presentation of media is usually interactive.

Challenges for Multimedia System

- Synchronization – Maintaining temporal relationships between different media types
- Sequencing – playing frames in correct order / time frame in video
- Distributed Network – Latency and Delay
- Continuous media types such as video need a lot of space to store and very high bandwidth to transmit.
- Multimedia involves many different research areas and needs more complex and more efficient algorithms and hardware platforms.

Component of Multimedia System

Now let us consider the Components (Hardware and Software) required for a multimedia system:

- **Capture devices** — Video Camera, Video Recorder, Audio Microphone, Keyboards, mice, graphics tablets, 3D input devices, tactile sensors, VR devices. Digitising Hardware
- **Storage Devices** — Hard disks, CD-ROMs, DVD-ROM, etc
- **Communication Networks** — Local Networks, Intranets, Internet, Multimedia or other special high speed networks.
- **Computer Systems** — Multimedia Desktop machines, Workstations, MPEG/VIDEO/DSP Hardware
- **Display Devices** — speakers, HDTV, SVGA, Hi-Resolution monitors, Colour printers etc.